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REVIEW

Neoadjuvant systemic therapy in geriatric breast cancer patients: a National Cancer Database (NCDB) analysis

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Abstract

Purpose Neoadjuvant systemic therapy (NAST) can be an effective treatment option for patients with HER2 + or triple negative breast cancer (TNBC). However, its use in geriatric patients is largely understudied. Our aim is to investigate the effect of NAST in both septuagenarians and octogenarians with HER2 + or TNBC to better understand its role in the geriatric patient population.

Methods We utilized the National Cancer Database (NCDB) to analyze female patients with HER2 + or TNBC between 70 and 89 years. We compared the baseline demographic and clinical characteristics of septuagenarians and octogenarians using mixed-effect modeling for continuous variables and conditional logistic regressions for categorical variables. Overall survival (OS) between several subgroups was compared based on a propensity score model. Kaplan–Meier method was used to calculate OS between the subgroups, and log-rank test was used to compare OS results.

Results A total of 16,443 patients met inclusion/exclusion criteria, of which 92.9% had infiltrative ductal carcinoma and 73.5% were TNBC. Most patients received NAST as a first course of therapy (58.8%). Septuagenarians were more likely to receive NAST (65.9%), whereas octogenarians were more likely to receive upfront surgical resection (67.7%). Our analysis demonstrated OS benefit with NAST among patients who received surgical resection. However, in patients who received NAST, decline during therapy was associated with a significantly poorer OS outcomes in general.

Conclusion When combined with surgical resection, NAST is an effective treatment option in both septuagenarians and octogenarians. Nonetheless, careful selection of NAST recipients in this population remains critical to optimize patient outcome.

Keywords Breast cancer · Geriatric · Neoadjuvant systemic therapy · Toxicity

Introduction

Breast cancer is diagnosed in approximately 13% of women in their lifetime, with the probability of diagnosis being highest in a woman's sixth decade [1]. However, the risk of death from breast cancer increases with increasing age of diagnosis [1]. With rapid growth of the aging population and 31% of breast cancer patients being above the age of 70, it is critical to optimize treatment for both short- and long-term care of geriatric breast cancer patients [1, 2].

Hormone receptor (HR) status and human epidermal growth factor receptor 2 (HER2) status are used to determine breast cancer subtype and inform treatment decisions [3]. Current treatment strategy often involves a combination of surgery, endocrine therapy, chemotherapy, and radiation, with the considerations for treatment not only being which therapies but also their sequence [3]. Neoadjuvant systemic therapy (NAST) is an established treatment option used to reduce tumor size for patients who desire more conservative surgery as well as to evaluate tumor biology early and tailor future treatment based on response to NAST [4]. NAST has also shown to be particularly useful in treating two of the more aggressive subtypes of breast cancer, triple-negative breast cancer (TNBC) and

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HER2 + breast cancer, which make up approximately 10% and 14% of breast cancer cases, respectively [1, 4]. However, the application of NAST in geriatric patients has been largely understudied, limiting our current understanding of its risks and benefits which must be carefully weighed on individual basis [5].

One major concern of NAST in geriatric patients is the increased risk of toxicity which can lead to dose reduction, treatment delays and omissions, or hospitalization [6]. Geriatric patients are also at increased risk of having comorbid conditions compared to non-geriatric patients [7, 8]. These comorbidities are not only associated with worse OS from breast cancer, but also with increased toxicity from chemotherapy [6, 9, 10]. On the other hand, evidence also suggests that about half of geriatric breast cancer patients are undertreated, in part due to providers' intention to avoid harm from treatment toxicities, resulting in decreased disease-specific survival [11, 12]. Although the data on NAST in geriatric breast cancer patients are limited, there are data to support pathologic complete response (pCR) rates to NAST in HER2 + geriatric breast cancer patients are similar to those of younger patients, and many patients had improved prognosis even without pCR [13].

Although there is evidence to support the use of NAST for TNBC and HER2 + breast cancer in non-geriatric patients, there is insufficient data regarding the risks and benefits of NAST in geriatric patients. Our aim of this analysis is therefore to investigate the effect of NAST on the treatments plans in geriatric breast cancer patients with TNBC or HER2 + breast cancer, and to investigate factors that may influence OS in these patients.

Methods

We utilized the National Cancer Database (NCDB) for breast cancer 2004–2017 for this analysis. We applied a series of inclusion/exclusion criteria as follows:

- Demographic criteria: only female patients, 70–89 years of age (i.e., septuagenarians and octogenarians), with unilateral breast cancer were selected.
- Histologic criteria: only patients with infiltrative ductal carcinoma (IDC) or infiltrative lobular carcinoma (ILC) were selected. Other histologies such as phyllodes tumors and breast sarcomas were excluded. Moreover, favorable IDC subtypes that are not usually recommended to undergo systemic therapy (cribriform, mucinous, and solid papillary) were removed from the dataset.
- Molecular criteria: only patients with positive HER2, whether HR positive (i.e., luminal B subtype) or HR negative (i.e., HER2 + subtype), and those with TNBC were included.
- Staging criteria: only patients with non-metastatic tumors > 1 cm in size (cT1c-T4) with a known clinical nodal status (cN0-N3) were included.
- Treatment criteria: to answer the clinical question of this work, we included patients who received NAST as a first course of therapy prior to surgical resection and those in whom NAST was omitted and underwent upfront surgical resection. Patients who were placed on neoadjuvant endocrine therapy as first course of treatment, patients with unknown status of NAST, and those with an unknown breast surgical intervention were excluded from the dataset. Moreover, we removed patients with no reported reason for omitting NAST or surgical resection.

After applying the inclusion/exclusion criteria, we also studied the group of patients who received NAST as first course of therapy to determine the proportion who medically declined during NAST and did not receive surgical resection. Likewise, we studied patients in whom NAST was omitted and underwent upfront surgical resection to determine the proportion that was deemed medically unfit for systemic therapy. Among patients who received NAST, we applied a backward stepwise logistic regression (entry condition, $p < 0.05$; removal condition, $p > 0.1$) to identify predictors of functional decline and omission of planned surgical resection.

We then divided the dataset into septuagenarian and octogenarian age groups. We compared the baseline demographic and clinical characteristics of these two groups using mixed-effect modeling for continuous variables and conditional logistic regressions for categorical variables.

Finally, we aimed to compare OS between several subgroups of patients as follows: (1) septuagenarians who had NAST and surgery (+ NAST + Surg) vs. those who had upfront surgery (− NAST + Surg); (2) octogenarians with + NAST + Surg vs. − NAST + Surg; (3) septuagenarians who had + NAST + Surg vs. those who declined during NAST and surgery was omitted (i.e., + NAST − Surg); and finally, (4) octogenarians who had + NAST + Surg vs. + NAST − Surg. These subgroup comparisons are summarized as:

(1) septuagenarians + NAST + Surg vs. − NAST + Surg,
(2) octogenarians + NAST + Surg vs. − NAST + Surg
(3) septuagenarians + NAST + Surg vs. + NAST − Surg, and
(4) octogenarians + NAST + Surg vs. + NAST − Surg.

To achieve this aim, we matched each pair of subgroups based on a propensity score model that takes into account the following variables: age, race, type of treating facility,

Charlson comorbidity index, laterality, receptor status, type of breast cancer, grade, clinical T stage, clinical N stage, and adjuvant endocrine therapy. For the latter two comparisons, we also included the type of NAST regimen that was offered (single agent vs. multi-agent). The match was performed based on a propensity score that was calculated from the regression model, and peers were matched 1:1 using the nearest neighbor methods with a caliper width of < 0.1. After matching, Kaplan–Meier method was used to calculate OS between the subgroups, and log-rank test was used to compare OS results.

Results

NCDB for breast cancer included 2,950,631 patients diagnosed between 2004 and 2017. After applying our inclusion/exclusion criteria, 16,443 patients were selected. Figure 1 is a flow diagram summarizing the selection steps.

The mean age of the selected population was 76.0 ± 4.8 years, and the majority were White (*N* = 13,097, 79.7%) and healthy with Charlson comorbidity score of 0 (*N* = 12,622, 76.8%). The most common histology was IDC (*N* = 15,281, 92.9%) and 12,090 (73.5%) cases were TNBC. Moreover, most of the patients had tumors ≤ 5 cm (*N* = 15,478, 94.1%) and were clinically node negative (*N* = 13,918, 84.6%). Regarding treatment choices, 9676 (58.8%) patients received NAST as first course of therapy, whereas in 6767 (41.2%) patients, NAST was omitted and surgery was offered as first course of therapy. Table 1 demonstrates the clinical and demographic characteristics of the selected patient population. Figure 2 shows the proportion of patients who declined on NAST as first course of therapy, and the proportion in whom NAST was omitted due to medical comorbidities.

Among patients who had NAST as first course of therapy (*N* = 9676), it was evident that the majority were septuagenarians (*N* = 8393, 86.7%), healthy with Charlson score 0 (*N* = 7528, 77.8%), about half had tumors ≥ cT2 (4160, 43%), and about quarter had node-positive disease (*N* = 2192, 22.7%). The mean time from diagnosis to initiation of therapy in this group of patients was

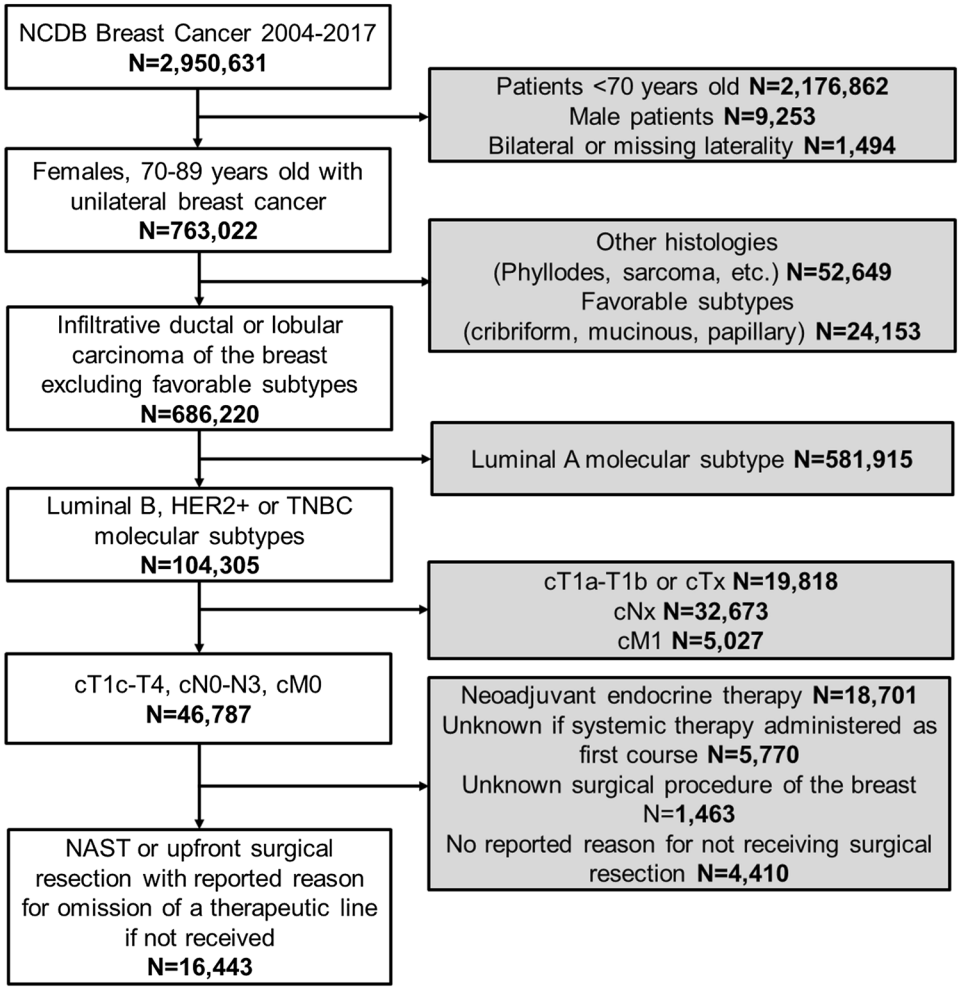


Fig. 1 Flow diagram of selection steps

Table 1 Demographic and clinical characteristics of the selected patient population

N		16,443
Age	Mean ± SD (median)	76.0 ± 4.8 (75)
Race	White	13,097 (79.7%)
	Black	2321 (14.1%)
	Other	1025 (6.2%)
Charlson Score	0	12,622 (76.8%)
	1	2794 (17.0%)
	2	714 (4.3%)
	3+	313 (1.9%)
Laterality	Right	8049 (49.0%)
	Left	8394 (51.0%)
Receptor status	HR + HER2 +	1295 (7.9%)
	HR- HER2 +	3058 (18.6%)
	TNBC	12,090 (73.5%)
Breast cancer type	IDC	15,281 (92.9%)
	ILC	1162 (7.1%)
Nottingham grade	Low	844 (5.1%)
	Intermediate	4088 (24.9%)
	High	10,094 (61.4%)
	Missing	1417 (8.6%)
Clinical T stage	cT1	10,906 (66.3%)
	cT2	4572 (27.8%)
	cT3	485 (2.9%)
	cT4	480 (2.9%)
Clinical N stage	cN0	13,918 (84.6%)
	cN1	1961 (11.9%)
	cN2	364 (2.2%)
	cN3	20 (1.2%)
Treatment received	+ NAST + Surg	9328 (56.7%)
	− NAST + Surg	6767 (41.2%)
	+ NAST − Surg	348 (2.1%)
Endocrine therapy	No	15,264 (92.8%)
	Yes	1179 (7.2%)

IDC: Infiltrative Ductal Carcinoma; *ILC*: Infiltrative Lobular Carcinoma; *HER*: Human Epidermal Growth Factor Receptor; *HR*: Hormone Receptor; *NAST*: Neoadjuvant Systemic Therapy; *SD*: Standard Deviation; Surg: Surgical Resection; *TNBC*: Triple Negative Breast Cancer

64.9 ± 37.7 days (median, 61), and 8538 patients (88.7%) had lumpectomies. 7515 (77.7%) of patients received adjuvant systemic therapy following surgical resection. Table 2 summarizes the characteristics of patients who had NAST as first course of therapy.

On the other hand, patients who had upfront surgical resection ($N = 6767$) were less likely to be septuagenarians ($N = 2689$, 60.3%), and more likely to have smaller tumors < 2 cm ($N = 5390$, 79.7%), and node-negative disease ($N = 6434$, 95.1%), and the majority underwent lumpectomy for their breast resection ($N = 6568$, 97.1%). Mean time between diagnosis and surgery was 37.1 ± 33.6 days (median, 30). Only 173 (2.6%) patients received adjuvant systemic therapy following surgical resection. Table 3 demonstrates the characteristics of patients who underwent surgical resection as first course of therapy.

Our logistic regression suggested that octogenarians were 6.85 times (vs. septuagenarians), Black patients were 2.19 times (vs. White), patients with Charlson score 3 were 2.13 times more likely (vs. Charlson score 0) to decline while on NAST and omit the planned surgical resection. Inversely, HR- HER2 + patients were less likely to decline during NAST (hazard ratio 0.65). Table 4 shows the final block of the multivariable logistic regression.

We divided the population into septuagenarian and octogenarian age groups. Differences were noted between the groups in race distribution, receptor status, breast cancer type, T stage, and N stage. Most importantly, a significant difference was noted in the treatment outlines where septuagenarians were more likely to receive NAST followed by surgical resection (65.9% vs. 28.1%; $p < 0.001$), whereas octogenarians were more likely to proceed with upfront surgical resection (32.7% vs. 67.7%; $p < 0.001$). Table 5 shows the comparative analysis between septuagenarians and octogenarians in the selected population.

Finally, to perform an adjusted OS analysis, we elected to divide the database into study groups as described in Methods to study whether NAST provides any survival benefit in different scenarios:

(1) septuagenarians + NAST + Surg vs. − NAST + Surg,
(2) octogenarians + NAST + Surg vs. − NAST + Surg
(3) septuagenarians + NAST + Surg vs. + NAST − Surg, and
(4) octogenarians + NAST + Surg vs. + NAST − Surg.

In each scenario, the study groups were propensity score matched to adjust for all clinical and demographic variables that could affect OS, especially T and N stages. We were able to match 3626 pairs in scenario 1, 1945 pairs in scenario 2, 2108 pairs in scenario 3, and finally 51 pairs of patients in scenario 4. Supplementary Tables 1–4 demonstrate the match process in scenarios 1–4, respectively, with comparative analysis of the study groups before and after the match. In sum, the match process produced well-balanced groups with adequate adjustment and resolution of all baseline differences between the study groups. Figure 3 illustrates the Kaplan–Meier plots for OS in matched subgroups of patients in the four scenarios.

Our Kaplan–Meier analysis demonstrated an OS benefit with NAST among patients who received surgical resection. In septuagenarians, median OS were not reached in either group and the 5-year absolute OS benefit was 6%

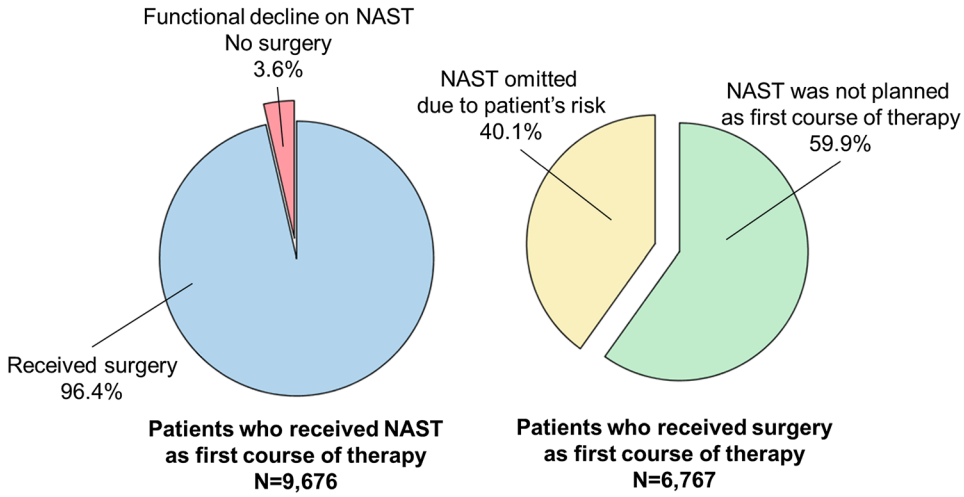


Fig. 2 Pie charts demonstrating NAST patients with the proportion who did not receive surgery due to medical decline while on NAST; and upfront surgery patients with the proportion who did not receive NAST due to medical risk factors as deemed by the treating physician

with + NAST + Surg vs. − NAST + Surg (86% vs. 80%; $p < 0.001$). In octogenarians, median OS with significantly improved with NAST (107.8 ± 4.7 vs. 80.7 ± 5.2 months; $p < 0.001$) and the 5-year absolute OS benefit was 10% (71% vs. 61%; $p < 0.001$).

In patients who received NAST as first course of therapy, decline during therapy was associated with a significantly poorer OS outcomes in general. In Septuagenarians, median OS was significantly longer in + NAST + Surg vs. + NAST − Surg patients (74.2 ± 7.1 vs. 28.3 ± 5.0 months; $p < 0.001$) with a 5-year OS difference of 26% (62% vs. 36%; $p < 0.001$). Similar results were seen among matched octogenarians in median OS (61.4 ± 10.3 vs. 16.2 ± 2.4 months; $p < 0.001$) and in 5-year OS (54% vs. 10%; $p < 0.001$).

Discussion

Our analysis suggests that when surgical resection is performed for TNBC and HER2 + breast cancers in geriatric patients, the addition of NAST confers an OS benefit for both septuagenarians and octogenarians. However, patients who declined during NAST and had their planned surgery omitted demonstrated very poor OS outcomes. The current body of evidence clearly supports the benefit of NAST for both HER2 + and TNBC patients, although the majority of studies on NAST do not specifically address the response of geriatric patients [5, 14, 15]. One study that has investigated NAST in this age group found that HER2 + patients > 65 years demonstrated pCR from NAST at rates as high as those < 40 years old [13]. However, the use of NAST in geriatric patients may also be associated with increased risk of toxicity [6, 16, 17]. While mitigating this increased risk of toxicity is critical in all patients, it may be vital in the octogenarians.

In the OS curves pertaining to patients who received surgical resection with or without NAST, there is in both instances an initial divergence followed by an equal and steady decline which suggests this difference in survival may be influenced by factors other than cancer progression such as medical comorbidities. A similar graph is seen in the OS curve of septuagenarians who received NAST with or without surgical resection, as the separation of curves remains mostly constant and ultimately reconverges. However, in the OS curve evaluating octogenarian recipients of NAST with or without surgical resection, there is an expanding difference for the first 76 months, suggesting the worse OS may be due to cancer progression rather than medical comorbidities. This is a critical difference from the septuagenarians, as it indicates that if toxicities from the NAST are significant enough to eliminate the option of surgical resection in octogenarians, then they may have worse OS. Similarly, compared to septuagenarians, octogenarians were almost seven times more likely to decline on NAST and therefore omit surgical resection. This importance of surgical resection of breast cancer has been identified previously, as surgical resection in octogenarians is associated with a significant benefit in OS irrespective of neoadjuvant or adjuvant therapies (HR 3.3 [3.18–3.46] $P < 0.001$) [18]. These findings emphasize that careful selection of NAST candidates, particularly in octogenarians, is critical to mitigate this risk and improve their overall survival.

Careful selection is likely why the majority of patients who were chosen for NAST were septuagenarians with a low Charlson score, as current recommendations are to consider fit geriatric patients for NAST but to recommend less fit geriatric patients for upfront surgery [19]. The fitness of these geriatric patients largely refers to their burden of comorbidities, and the increased incidence of comorbidities with increased age aligns with the favored use of NAST in septuagenarians as compared

Table 2 Description of patient who had NAST as first course of therapy

N		9676
Age category	Septuagenarians	8393 (86.7%)
	Octogenarians	1283 (13.3%)
Race	White	7569 (78.2%)
	Black	1495 (15.5%)
	Other	612 (6.3%)
Charlson Score	0	7528 (77.8%)
	1	1610 (16.6%)
	2	381 (3.9%)
	3 +	157 (1.6%)
Laterality	Right	4743 (49.0%)
	Left	4933 (51.0%)
Receptor status	HR + HER2 +	666 (6.9%)
	HR- HER2 +	2189 (22.6%)
	TNBC	6821 (70.5%)
Breast cancer type	IDC	9037 (93.4%)
	ILC	639 (6.6%)
Nottingham grade	Low	322 (3.3%)
	Intermediate	2109 (21.8%)
	High	6368 (65.8%)
	Missing	877 (9.1%)
Clinical T stage	cT1	5516 (57.0%)
	cT2	3307 (34.2%)
	cT3	410 (4.2%)
	cT4	443 (4.6%)
Clinical N stage	cN0	7484 (77.3%)
	cN1	1695 (17.5%)
	cN2	315 (3.3%)
	cN3	182 (1.9%)
NAST regimen	Single agent	1242 (12.8%)
	Multi-agent	8183 (84.6%)
	Missing	251 (2.6%)
Days from diagnosis to initiation of therapy	Mean ± SD, Median	64.9 ± 37.7, 61
No surgery (declined performance)		348 (3.6%)
pCR (ypT0/Tis ypN0)		5422 (56.0%)
Breast surgery	No surgery	348 (3.6%)
	Lumpectomy	8583 (88.7%)
	Mastectomy	745 (7.7%)
Definitive axillary surgery	SLNB	4977 (51.4%)
	ALND	2443 (25.2%)
	Missing	2256 (23.4%)
Adjuvant systemic therapy		7515 (77.7%)
Endocrine therapy		604 (6.2%)

IDC: Infiltrative Ductal Carcinoma; *ILC*: Infiltrative Lobular Carcinoma; *HER*: Human Epidermal Growth Factor Receptor; *HR*: Hormone Receptor; *NAST*: Neoadjuvant Systemic Therapy; *pCR*: Pathologic Complete Response; *SD*: Standard Deviation; Surg: Surgical Resection; *TNBC*: Triple Negative Breast Cancer

to octogenarians [8]. However, while the patients who receive NAST may be more fit in terms of comorbidities, they are also more likely to have greater disease burden. The patients who had upfront surgery were more likely to have smaller tumors and node-negative disease, and only 2.6% of these patients required adjuvant systemic therapy. This 2.6% is in stark contrast to the 77.7% of

Table 3 Description of patients who had upfront surgery without NAST

N		6767
Age category	Septuagenarians	4078 (60.3%)
	Octogenarians	2689 (39.7%)
Race	White	5528 (81.7%)
	Black	826 (12.2%)
	Other	413 (6.1%)
Charlson Score	0	5094 (75.3%)
	1	1184 (17.5%)
	2	333 (4.9%)
	3 +	156 (2.3%)
Laterality	Right	3306 (48.9%)
	Left	3461 (51.1%)
Receptor status	HR + HER2 +	629 (9.3%)
	HR − HER2 +	869 (12.8%)
	TNBC	5269 (77.9%)
Breast cancer type	IDC	6244 (92.3%)
	ILC	523 (7.7%)
Nottingham grade	Low	522 (7.7%)
	Intermediate	1979 (29.2%)
	High	3726 (55.1%)
	Missing	540 (8.0%)
Clinical T stage	cT1	5390 (79.7%)
	cT2	1265 (18.7%)
	cT3	75 (1.1%)
	cT4	37 (0.5%)
Clinical N stage	cN0	6434 (95.1%)
	cN1	266 (3.9%)
	cN2	49 (0.7%)
	cN3	18 (0.3%)
Days from diagnosis to initiation of therapy	Mean ± SD, Median	37.1 ± 33.6, 30
No NAST due to patient's risk		2706 (40.1%)
Breast surgery	Lumpectomy	6568 (97.1%)
	Mastectomy	199 (2.9%)
Definitive axillary surgery	SLNB	3563 (52.7%)
	ALND	1020 (15.1%)
	Missing	2184 (32.3%)
Adjuvant systemic therapy		173 (2.6%)
Endocrine therapy		575 (8.5%)

IDC: Infiltrative Ductal Carcinoma; *ILC*: Infiltrative Lobular Carcinoma; *HER*: Human Epidermal Growth Factor Receptor; *HR*: Hormone Receptor; *NAST*: Neoadjuvant Systemic Therapy; *pCR*: Pathologic Complete Response; *SD*: Standard Deviation; *Surg*: Surgical Resection; *TNBC*: Triple Negative Breast Cancer

patients who received NAST that also required adjuvant systemic therapy.

Unfortunately, NCDB does not include a specific report on the systemic therapy regimens. Therefore, whether adjuvant systemic therapy was part of an original "sandwich" plan or a subsequent decision based on final pathology remains unclear and is admittedly a weakness of this data registry. However, we examined our database more closely in attempt to understand the possible indications of adjuvant therapies in the 77.7% of NAST cases. Of the 7515 patients who received NAST, surgery, and adjuvant systemic therapy, 2279 (30.3%) were HER2 +. In that particular scenario, adjuvant systemic therapy likely indicates the receipt of adjuvant Trastuzumab ± Pertuzumab since NCDB does not specify the type of systemic therapy regimen and reports on chemotherapy and targeted therapies under the same variable. Of the 5236 patients (69.7%) who had TNBC, of whom 2407 did not achieve pCR on final pathology, this subgroup may have received adjuvant systemic therapy due to the residual disease after NAST. Finally, in the remaining 2829 TNBC patients who achieved pCR and still received adjuvant systemic therapies, we presume that this would have been a planned "sandwich" regimen, especially when learning that in this specific subgroup, 2623 (92.7%) were septuagenarians and only 206 (7.3%) were octogenarians, further suggesting that this is a pre-planned regimen by the treating oncologist in a relatively younger and healthier patient subgroup.

Despite the increased rates of patients requiring adjuvant systemic therapy, the patients who received both surgery and NAST had improved OS compared to those who had upfront surgery, further supporting the use of NAST in fit geriatric

Table 4 Results of the final block of the backward stepwise multivariable logistic regression for predictors of decline during NAST and omission of surgical resection among patients who received NAST as first course of therapy ($N = 9676$) in the selected patient population

		Hazard Ratio [95% CI]	*p*
Age group	Septuagenarian	Referent	
	Octogenarian	6.85 [5.46–8.59]	**<0.001***
Race	White	Referent	
	Black	2.19 [1.68–2.84]	**< 0.001***
	Other	1.48 [0.92–2.24]	0.079
Charlson score	0	Referent	
	1	0.89 [0.66–1.20]	0.456
	2	1.45 [0.89–2.32]	0.128
	3 +	2.13 [1.15–3.98]	**0.015***
Receptor status	HR + HER2 +	0.85 [0.63–1.11]	0.105
	HR- HER2 +	0.65 [0.48–0.89]	**0.007***
	TNBC	Referent	
NAST regimen	Single agent	Referent	
	Multi-agent	0.73 [0.54–1.05]	0.063
	Missing	0.69 [0.48–1.10]	0.089

HER: Human Epidermal Growth Factor Receptor; *HR*: Hormone Receptor; *NAST*: Neoadjuvant Systemic Therapy

Results in bold are statistically significant

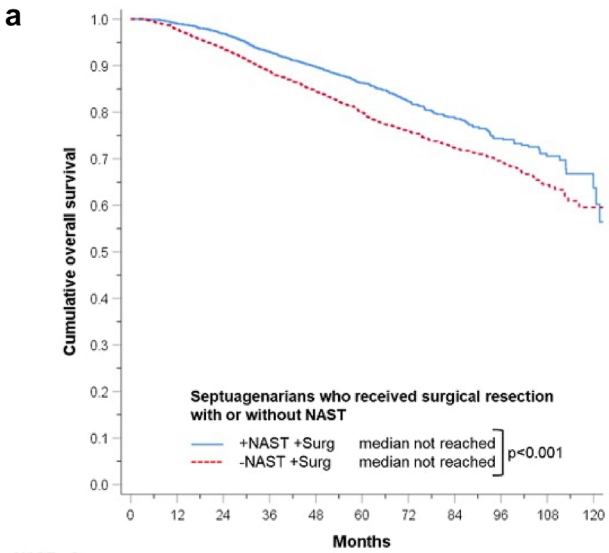
Table 5 Comparison of baseline characteristics between septuagenarian and octogenarian patients in the selected population

		Septuagenarians (70–79 years)	Octogenarians (80–89 years)	*P* value
N		12,471	3,972	
Age	Mean ± SD	73.7 ± 2.7	83.2 ± 2.5	< 0.001*
Race	White	9809 (78.7%)	3288 (82.8%)	< 0.001*
	Black	1873 (15.0%)	448 (11.3%)	
	Other	789 (6.3%)	236 (5.9%)	
Charlson score	0	9615 (77.1%)	3007 (75.7%)	0.331
	1	2084 (16.7%)	710 (17.9%)	
	2	537 (4.3%)	537 (4.3%)	
	3 +	235 (1.9%)	78 (2.0%)	
Laterality	Right	6096 (48.9%)	1,953 (49.2%)	0.752
	Left	6375 (51.1%)	2,019 (50.8%)	
Receptor status	HR + HER2 +	935 (7.5%)	360 (9.1%)	0.005*
	HR − HER2 +	2,41 (18.8%)	717 (18.1%)	
	TNBC	9195 (73.7%)	2895 (72.9%)	
Breast cancer type	IDC	11,652 (93.4%)	3629 (91.4%)	< 0.001*
	ILC	819 (6.6%)	343 (8.6%)	
Nottingham grade	Low	626 (5.0%)	218 (5.5%)	0.074
	Intermediate	3063 (24.6%)	1025 (25.8%)	
	High	7677 (61.6%)	2417 (60.9%)	
	Missing	1105 (8.9%)	312 (7.9%)	
Clinical T stage	cT1	8394 (67.3%)	2512 (63.2%)	< 0.001*
	cT2	3410 (27.3%)	1162 (29.3%)	
	cT3	367 (2.9%)	118 (3.0%)	
	cT4	300 (2.4%)	180 (4.5%)	
Clinical N stage	cN0	10,599 (85.0%)	3319 (83.6%)	< 0.001*
	cN1	1489 (11.9%)	472 (11.9%)	
	cN2	245 (2.0%)	119 (3.0%)	
	cN3	138 (1.1%)	62 (1.6%)	
Treatment received	+ NAST + Surg	8213 (65.9%)	1115 (28.1%)	< 0.001*
	− NAST + Surg	4078 (32.7%)	2689 (67.7%)	
	+ NAST − Surg	180 (1.4%)	168 (4.2%)	
Endocrine therapy	No	11,616 (93.1%)	3648 (91.8%)	0.006*
	Yes	855 (6.9%)	324 (8.2%)	

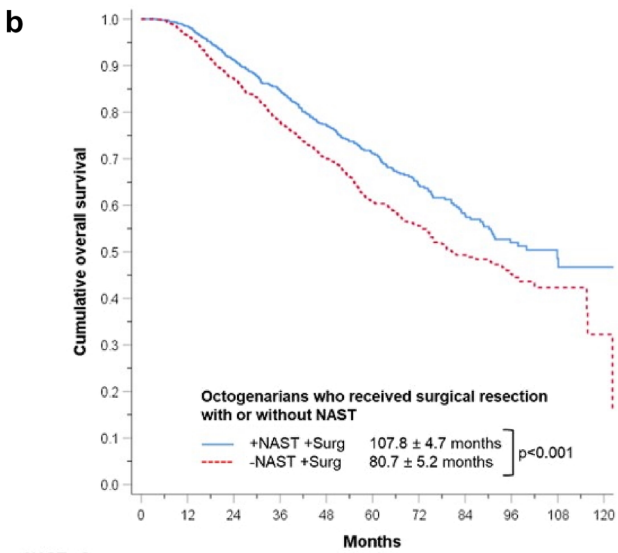
IDC: Infiltrative Ductal Carcinoma; *ILC*: Infiltrative Lobular Carcinoma; *HER*: Human Epidermal Growth Factor Receptor; *HR*: Hormone Receptor; *NAST*: Neoadjuvant Systemic Therapy; *SD*: Standard Deviation; *Surg*: Surgical Resection; *TNBC*: Triple Negative Breast Cancer. *Statistically significant

candidates. This aligns with a previous investigation that evaluated 13 comorbid conditions in relation to geriatric breast cancer patients and found that patients with a comorbidity and lower stage breast cancer had worse OS than patients with no comorbidities and higher stage breast cancer [10]. It is also consistent with the finding that patients with a Charlson score of 3 were more likely to decline on NAST and therefore omit the planned surgical resection compared to those with Charlson score of 0.

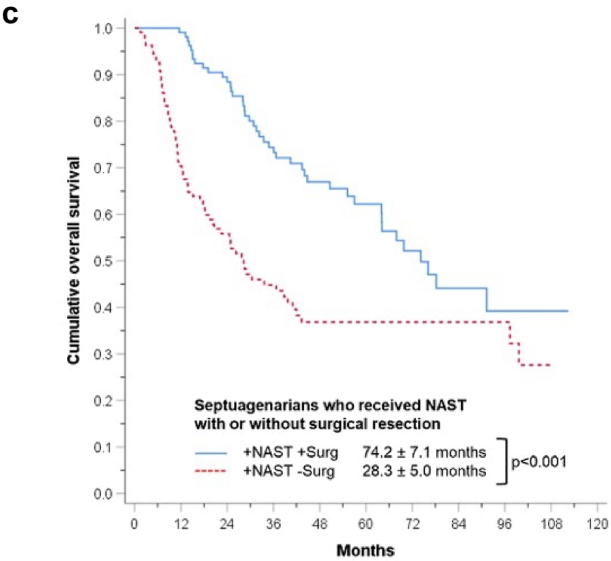
These data all suggest that NAST can improve OS if given to the right candidates. Although careful selection for NAST candidacy can be challenging, there are several tools that have been created to aid in this selection. The Chemotherapy Risk Assessment Scale for High-Age Patients (CRASH) score is designed to evaluate and stratify geriatric patients based on their risk of toxicity from chemotherapy [20]. Similarly, the Cancer and Aging Research Group-Breast Cancer (CARG-BC) score has been validated to predict chemotherapy toxicity specifically in geriatric breast cancer patients [21]. Another useful assessment is the Geriatric Assessment (GA) tool, which uses a multidisciplinary approach to evaluate the patient's functional status, medical comorbidities, and cognition, among other factors, to predict chemotherapy toxicity and mortality [22, 23]. Utilization of these tools can help determine which patients are suitable



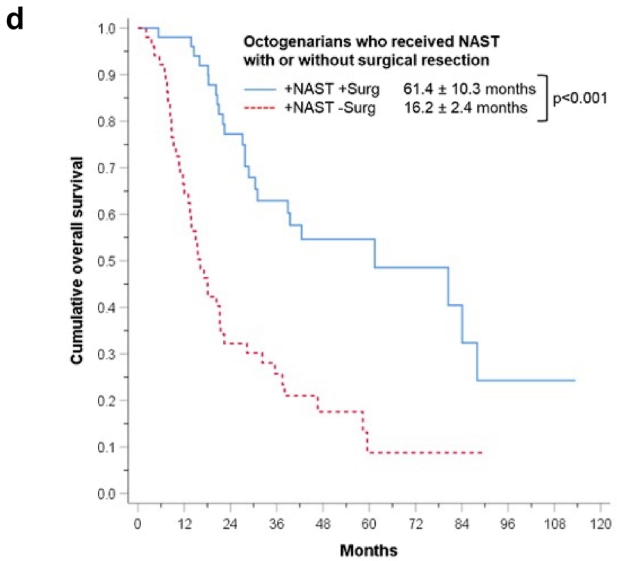
	0	12	24	36	48	60	72	84	96	108	120
+NAST +Surg											
N entering	3,626	3,540	3,219	2,546	1,956	1,400	949	602	325	127	
N withdrawing	52	246	554	513	490	398	313	250	188	101	
N exposed to risk	3,600	3,417	2,942	2,289	1,711	1,201	792	477	231	76	
N of events	34	75	119	77	66	53	34	27	10	5	
Cumulative survival	0.99	0.97	0.93	0.90	0.86	0.83	0.97	0.75	0.71	0.67	
-NAST +Surg											
N entering	3,626	3,459	3,093	2,457	1,957	1,496	1,078	709	428	151	
N withdrawing	87	224	490	391	366	351	326	259	256	119	
N exposed to risk	3,582	3,347	2,848	2,261	1,774	1,320	915	579	300	91	
N of events	80	142	146	109	95	67	43	22	21	6	
Cumulative survival	0.98	0.94	0.89	0.85	0.80	0.76	0.72	0.70	0.65	0.61	



	0	12	24	36	48	60	72	84	96	108	120
+NAST +Surg											
N entering	945	914	779	598	429	301	207	131	75	26	
N withdrawing	17	70	129	123	97	69	59	45	46	17	
N exposed to risk	936	879	714	536	380	266	177	108	52	17	
N of events	14	65	52	46	31	25	17	11	3	1	
Cumulative survival	0.99	0.91	0.85	0.77	0.71	0.64	0.58	0.52	0.49	0.46	
-NAST +Surg											
N entering	945	893	744	574	415	275	168	114	61	22	
N withdrawing	19	66	96	107	93	88	37	46	36	18	
N exposed to risk	935	860	696	520	368	231	149	91	43	13	
N of events	33	83	74	52	47	19	17	7	3	2	
Cumulative survival	0.96	0.87	0.78	0.70	0.61	0.56	0.50	0.46	0.43	0.36	



	0	12	24	36	48	60	72	84	96	108	120
+NAST +Surg											
N entering	108	105	87	65	49	35	23	10	6	4	
N withdrawing	2	7	9	10	11	7	10	3	2	4	
N exposed to risk	107	101	82	60	43	31	18	8	5	2	
N of events	1	11	13	6	3	5	3	1	0	0	
Cumulative survival	0.99	0.89	0.74	0.67	0.62	0.52	0.44	0.39	0.39	0.39	
+NAST -Surg											
N entering	108	76	53	37	21	14	11	10	8	1	
N withdrawing	0	8	6	10	7	3	1	2	5	1	
N exposed to risk	108	72	50	32	17	12	10	9	5	0	
N of events	32	15	10	6	0	0	0	0	2	0	
Cumulative survival	0.70	0.56	0.45	0.36	0.36	0.36	0.36	0.36	0.23	0.23	



	0	12	24	36	48	60	72	84	96	108	120
+NAST +Surg											
N entering	51	49	36	25	14	12	8	5	3	1	
N withdrawing	1	3	5	8	2	3	2	0	2	1	
N exposed to risk	50	47	33	21	13	10	7	5	2	1	
N of events	1	10	6	3	0	1	1	2	0	0	
Cumulative survival	0.98	0.77	0.64	0.54	0.54	0.49	0.42	0.25	0.25	0.25	
+NAST -Surg											
N entering	51	33	16	11	5	2	1	1	---	---	
N withdrawing	1	0	2	3	1	1	0	1	---	---	
N exposed to risk	50	33	15	9	4	1	1	1	---	---	
N of events	17	17	3	3	2	0	0	0	---	---	
Cumulative survival	0.66	0.32	0.26	0.18	0.10	0.10	0.10	0.10	---	---	

Fig. 3 Kaplan–Meier plots for overall survival in matched subgroups of patients in the following scenarios, **a** septuagenarians + NAST + Surg vs. − NAST + Surg, **b** octogenarians + NAST + Surg vs. − NAST + Surg, **c** septuagenarians + NAST + Surg vs. + NAST − Surg, and **d** octogenarians + NAST + Surg vs. + NAST − Surg

candidates for NAST and may result in better OS for geriatric breast cancer patients.

Other notable findings included that HER2 + patients were less likely to decline during NAST than TNBC. This could be explained by either disease progression, or it may be due to differences in the NAST regimen used for these patients [24, 25]. Unfortunately, due to the limitations of the NCDB, there is insufficient information to adequately explain this finding. It was also found that Black women were more than twice as likely was White women to decline on NAST and therefore omit the planned surgical resection. Racial disparities in breast cancer have been well established, yet the cancer death rate remains 41% higher in non-Hispanic Black women than in non-Hispanic White women [26]. This disparity is likely multifactorial, with some factors being the increased incidence of comorbidities, later stage of diagnosis, unfavorable tumor characteristics, and barriers to adequate healthcare [26–28]. Multidisciplinary research and interventions are needed to address this gap in NAST decline and in breast cancer survival.

Since our study was conducted using the NCDB, we acknowledge that it may have several shortcomings most of which are inherent to the use of a national registry and its associated data loss and misreporting. Most importantly, the NCDB does not provide a detailed report on the specific regimens or toxicity of NAST, which limits our ability to conclude the reasons of functional decline and omission of surgical resection in patients who were started on NAST. Moreover, the NCDB does not report data on disease progression and recurrence, limiting the survival outcomes to OS analysis. As discussed above, patients' survival, especially in the geriatric population, is multifactorial and using the available data, we are not able to deduce whether the reported survival outcomes are disease-specific or attributed to patients' comorbidities.

However, we attempted to mitigate for these shortcomings by applying a strict propensity score matching including septuagenarians and octogenarians as separate age groups with acceptable adjustment for all demographic and clinical variables, including comorbidity scores, and treatment sequences. We believe that this analysis provides an idea about national trends of breast cancer management in geriatric patients. Our aim in this study was not to support one treatment sequence over another since these conclusions can only be drawn from well-designed and well-controlled clinical trials. Rather, we still believe that the treating physicians' clinical judgment remains the pillar upon which such decisions should be made, especially in the challenging geriatric population. From analyzing the outcomes on the large scale, we only aim to inform the decision of treatment modalities and sequence for geriatric breast cancer patients and raise the awareness about the potential scenarios and their associated outcomes.

Conclusion

Overall, these data illustrate the favorable survival benefit of NAST in geriatric breast cancer patients when combined with surgical resection, while also reinforcing the necessity of careful selection for which patients receive NAST. This is particularly true in octogenarians, as the use of NAST comes with the risk of worse OS if the patient's health declines to the point that they can no longer receive surgery. Although careful selection of recipients for NAST is not a novel concept, these data provide tangible evidence of its importance in geriatric breast cancer patients with TNBC or HER2 + breast cancer.

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Author contributions All authors contributed to the study conception and design. LB was involved in analysis, writing, and edits. SN was involved in statistical analysis, writing, and edits. CS and CL were involved in interpretation, revision, and review. MCL was involved in review of results, editing of manuscript, and review. The first draft of the manuscript was written by LB and SN, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The datasets analyzed during the current study are from the NCDB, a national registry. The American College of Surgeons (ACS) grants access to accredited Commission on Cancer (CoC) institutions via a standardized application process. Sharing these data with third parties is prohibited per the agreement with ACS upon granting the requested data file. Please refer to the Fellow of the American College of Surgeons (FACS) website to review the legal aspects and privacy policy related to this database.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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